

Directions. Complete all the problems. Your solutions to the problems should be submitted to Gradescope before the indicated due date above. Please follow rules regarding Gradescope submission as described in the syllabus. If you use any outside source in your solutions, you must cite the source.

Problem 1. Consider the differential equation

$$y' = y^2.$$

- (a) Is this a linear ODE? Explain your answer.
- (b) Find the general solution for this ODE using the fact that it is a *separable equation*.
- (c) Solve the IVP with initial condition $y(0) = 2$.

Problem 2. Consider the linear ODE

$$(*) \quad y' - 2y - t = 0.$$

- (a) Write down the corresponding homogeneous version of $(*)$ and find its general solution.
- (b) Check that

$$y(t) = -\left(\frac{t}{2} + \frac{1}{4}\right)$$

is a solution to $(*)$.

- (c) Write down the general solution to $(*)$.

Problem 3. Find the general solution to the following differential equations.

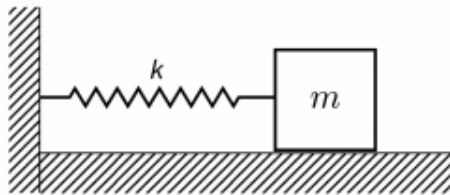
- (a) $2y'' - 4y' + 8y = 0$
- (b) $y'' - 4y' + 4y = 0$

(c) $y'' - 9y' + 20y = 0$

Problem 4. Consider a block with mass m attached to a spring that is anchored to a wall. This system is oriented horizontally and its location at time t (measured in seconds) is $x(t)$. Suppose that we have chosen a coordinate system so that at rest the mass' position is at $x = 0$. The horizontal motion of the spring mass is governed by Hooke's law,

$$mx'' = -kx,$$

where $k > 0$ is some physical constant.



- (a) We stretch the mass to position $x = 3$ and the release it, so that the initial conditions are $x(0) = 3$ and $x'(0) = 0$. Find the trajectory $x(t)$ (i.e., the solution to the IVP).
- (b) In the trajectory from (a), what is the farthest distance away from position $x = 0$ that the mass obtains?
- (c) What is the period of the oscillation (the amount of time it takes for the mass to return to its original position)? What is the frequency (number of oscillations in 1 second of time)?

Problem 5. Define f by

$$f(t) = \begin{cases} 0 & \text{if } t \leq 0 \\ t & \text{if } 0 \leq t \leq 1 \\ 1 & \text{if } 1 \leq t \end{cases}$$

Compute $L[f]$.

Problem 6. The hyperbolic cosine and sine functions are defined by

$$\cosh(t) = \frac{e^t + e^{-t}}{2}, \quad \sinh(t) = \frac{e^t - e^{-t}}{2}$$

For any constant a , compute $L[\cosh(at)]$ and $L[\sinh(at)]$ without integrating.

Problem 7. Show how to derive an expression for $L[y''']$ in terms of $L[y]$.

Problem 8. Find a function $f(t)$ which satisfies

$$L[f](p) = \frac{4-p}{(p-1)(p+2)}$$

Hint. Begin by using partial fraction decomposition

$$\frac{4-p}{(p-1)(p+2)} = \frac{A}{p-1} + \frac{B}{p+2}.$$

Problem 9. Use Laplace transforms to solve the IVP

$$y' + 2y + 2 = 0, \quad y(0) = 0.$$